



Aitiome

A mechanistic reasoning engine for the environmental exposome of neurodegeneration.

It recovers the real ones, resolves to the real neurons, and isn't fooled by the imposters.

`aitiome.fly.dev` / Built with Claude: Life Sciences

What causes sporadic Parkinson's and Alzheimer's?

The large majority of Parkinson's and Alzheimer's is **sporadic** - not inherited. The leading suspect for the rest is the **environmental exposome**: the pesticides, metals, and industrial chemicals we're exposed to over a lifetime. In 2024 the field's leaders (Miller, Barouki, Samieri; *Nature Neuroscience*) called for AI that connects chemicals to disease *mechanism*.

Which chemicals actually drive neurodegeneration, and through what biological pathway? That is the open question this project takes on.

Bioactive is not the same as neurotoxic

Tens of thousands of chemicals are **bioactive** - they light up assays. Almost none of them cause neurodegeneration. The entire difficulty is telling a true neurotoxicant apart from a compound that is merely bioactive.

This is the exact failure mode of activity-based screening: it flags everything that *does something*. Beat that, and you have something real. Fail it, and you have a watchlist no scientist can trust.

The whole project is built to win that one discrimination.

A discovery engine for novel drivers

The original pitch: an AI that scans the environmental chemical universe and surfaces *novel* candidate drivers of neurodegeneration, validated by recovering the known neurotoxicants.

Before writing code, we did a multi-round data reconnaissance to check whether that was actually possible on public data. It changed the project - and that is the interesting part.

The evidence redirected us

FINDING 1

No shippable unsupervised-discovery signal exists for this chemical class on public data. Seven independent axes tested; all coverage-killed or confounder-killed.

FINDING 2

The discriminating signal is **curated mechanism**, not assay activity. Bioactivity scores **worse than chance** against the decoys.

So we built the thing the data supports - and made that discipline the contribution.

Three co-equal goals

01

Validation that works

Recover the known neurotoxicants on the endorsed pathway; reject the adversarial decoys.

02

A signature visualization

The recovery-and-specificity reveal resolving into the vulnerable dopaminergic neurons.

03

Calibrated framing

Confidence tiers on every result; the discovery limits shown, not hidden.

The recovery rule

```
positive ⇔ ( CTD curated Parkinson's DirectEvidence ) OR ( registered neuro-  
AOP stressor )
```

Curated signals are diagnostic.

Two independent curation efforts. A fixed logical rule, no fitted parameters to overfit.

Assay activity is corroboration only.

Anti-diagnostic here - the engine **never** gates a positive call on bioactivity.



● rotenone Resolves into the SOX6 / AGTR1 vulnerable dopaminergic neurons

THE SIGNATURE VISUAL

The endorsed cascade, grounded, resolving into the vulnerable neurons

MIE complex-I inhibition → mitochondrial dysfunction → nigrostriatal dopaminergic degeneration → parkinsonian deficits. Grounded in MitoCarta Complex-I subunits and the Kamath SOX6/AGTR1 vulnerable dopaminergic neurons.

THE RESULT, ON THE RECONNAISSANCE GROUND TRUTH

Zero errors

13/13

known neurotoxicants recovered

15/15

negatives correctly rejected

6/6

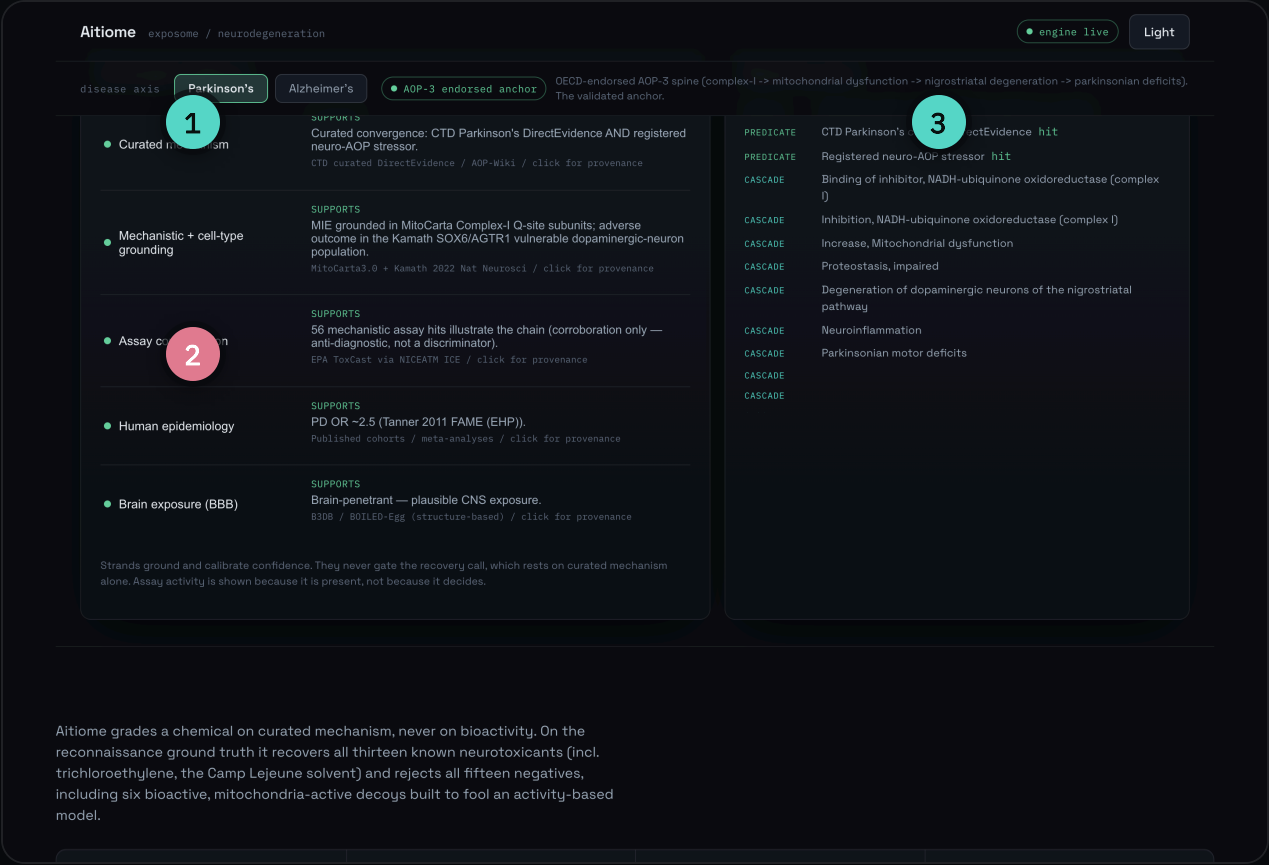
bioactive decoys not fooled

0

false positives + false negatives

13 Parkinson's positives (6 assay-recovered + 7 curated-anchored, incl. trichloroethylene / Camp Lejeune) - 15 negatives, including 6 mitochondria-active adversarial decoys built to fool an activity model.

Convergent evidence, and the pathway reconstructed



1 Five convergent evidence strands, each with provenance

2 Assay activity is corroboration only - never a discriminator

3 The endorsed AOP, reconstructed edge by edge

It is not fooled by the imposters

Aitiome

exposome / neurodegeneration

engine live

Light

disease axis

Parkinson's

Alzheimer's

AOP-3 endorsed anchor

OECD-endorsed AOP-3 spine (complex-I -> mitochondrial dysfunction -> nigrostriatal degeneration -> parkinsonian deficits). The validated anchor.

Every promising and similarity axis raised here because the decoys are genuinely bioactive, even mitochondria-active. Aitiome withholds the call anyway, because it reasons on curated mechanism. Pick a decoy: the left shows what fools an activity model; the right shows why the engine does not.

troglitazone

prochloraz

propiconazole

simvastatin

fenofibrate

warfarin

WHAT FOOLS AN ACTIVITY MODEL

Mitochondrial assays

0

Membrane potential (MMP)

1

Oxidative stress

1

Mechanistic assays (total)

6

ToxCast active (all)

26

Bioactive, and often mitochondria-active. On multi-assay fingerprint similarity the decoys collapse into the positives (0.53), which is why bioactivity cannot be the discriminator.

WHY AITIOME DOES NOT

3 independent lines

Curated mechanism

No curated Parkinson's DirectEvidence and not a registered neuro-AOP stressor.

Pharmacovigilance (FAERS)

Zero parkinsonism signal across 126794 FAERS reports (ROR 0.5, not elevated).

Brain exposure (BBB)

Low brain exposure (BBB-negative): an orthogonal pharmacokinetic reason it cannot be a driver.

The call rests on curated mechanism. Where the decoy is also non-penetrant or shows no pharmacovigilance signal, those are independent confirmations, not the reason.

THE FALSIFICATION

Why not just use bioactivity?

Computed live from the validation set. If activity could do the job, some bioactivity signal would separate the real neurotoxicants from the bioactive decoys. None does: every one is at or below chance against the decoys, while the curated rule is perfect on the same set.

BIOACTIVITY VS. THE 6 DECOYS (AIIROC)

0.5 = a chance

Why not just use bioactivity?

If activity could do the job, some signal would separate the 13 positives from the 6 decoys. None does - every one is at or below chance (0.5).



Curated rule on the same set: **perfect (0 errors)**. The decoys are, if anything, more bioactive than the real neurotoxicants.

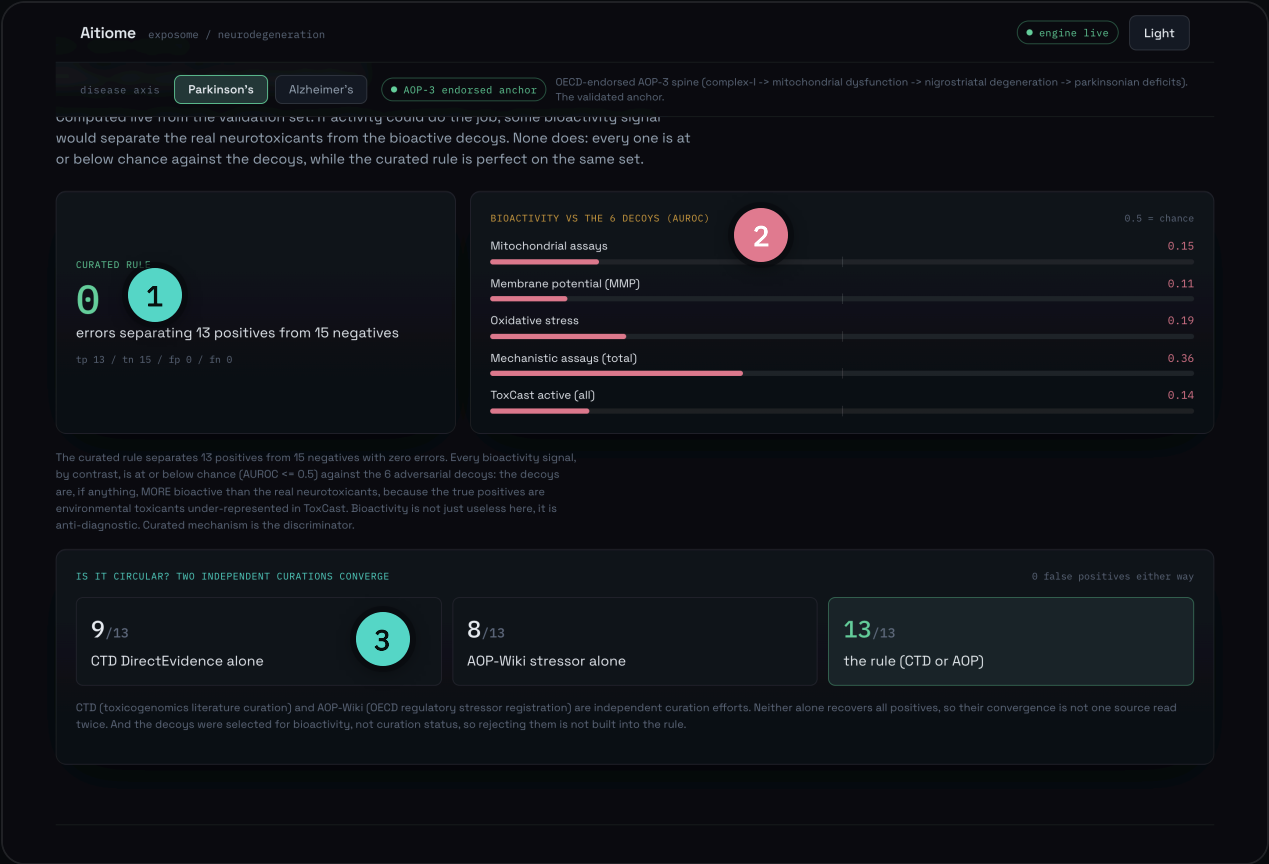
Prior chemical-disease ML reports 90%+ AUROC - but on drug-keyed, well-covered space, never adversarial environmental decoys. ToxCast covers only ~40% neural-relevant targets (Mack 2024): the structural reason activity fails here.

Is it circular? Two independent curations converge

$$\begin{array}{ccccc} 9/13 & + & 8/13 & \rightarrow & 13/13 \\ \text{CTD DirectEvidence alone} & & \text{AOP-Wiki stressor alone} & & \text{the rule (CTD or AOP)} \end{array}$$

CTD (toxicogenomics literature) and AOP-Wiki (OECD regulatory) are independent efforts; neither alone suffices, so this is not one source read twice. Both are **0/15** false-positives. And the decoys were selected for bioactivity, not curation status - so rejecting them is not baked in.

The whole argument, running in the product



1 Curated rule: zero errors on the set

2 Every bioactivity signal at or below chance

3 Circularity answered: two independent curations converge

Where AI-driven discovery works, and where it does not

LINCS L1000 COVERAGE-KILLED	EPA HTr COVERAGE-KILLED	EPA HTPP COVERAGE-KILLED	Structure / QSAR CONFOUNDER-KILLED
ToxCast fingerprint CONFOUNDER-KILLED	ComptoxAI / AlzKB COVERAGE-KILLED	Neural-specific subset QUALIFIED LEAD ●	Boltz-2 Q-site CONDITIONAL LEAD ●

Seven axes coverage- or confounder-killed. Two unproven leads remain (reported with explicit N). Discovery is a map, never a predictor.

Alzheimer's — calibrated below Parkinson's

THE AD ARM

The identical curated-diagnostic engine, anchored on the
endorsed AOP-12/48

(aging neurodegeneration + memory; registered stressor: lead) with a non-endorsed Tau/amyloid overlay. Recovers curated AD-linked chemicals (DDE, cadmium, lead), rejects the imposters, and resolves into disease-associated microglia (TREM2/APOE).

THE DECOYS ARE THE TREATMENTS

The compounds most active on AD assays are the
anti-amyloid
drugs

(donepezil, methylene blue) and dietary
polyphenols

(curcumin, EGCG). An activity model would flag the cure. Aitiome does not - no curated causation.

Same rigor; confidence calibrated to the evidence.

The same method, calibrated to the evidence.

dimension	Parkinson's	Alzheimer's
Scaffold (OECD)	AOP-3 endorsed; EFSA-expanding complex I-IV family	AOP-12/48 endorsed + non-endorsed overlay
Curated recovery	13/13, 0 err	12/12, 0 err
Predictive power / falsification	quantified: activity at or below chance	qualitative; assay-AUROC pending data
Circularity defense	two curations converge (9/13 + 8/13)	leans on CTD alone (~11/12)
Human epidemiology	18/16 + 10/5 + 10/5 = 43/26	22/22 + 11/12 + 10/10 = 43/44

Shared: KE-188 neuroinflammation bridges both AOPs; lead is positive for both; mitochondrial dysfunction is an unlinked gap, shown not hidden.

The field predicts. We validate.

The landscape

Graph-AI (PROTON, 2025) and read-across (GenRA) predict *novel* toxic chemicals from structure and bioactivity; a wave of LLM-for-AOP work is emerging.

Aitiome's inversion

Recover the *known* neurotoxicants on the endorsed scaffold, grade only on curated evidence, and prove via adversarial falsification that bioactivity is anti-diagnostic - the validating counterweight, and a direct answer to the hallucination warnings (ToxReason, ES&T 2024) the LLM-AOP wave raises.

We found no prior adversarial mito-active-decoy neurodegeneration benchmark.

Convergent with the mechanistic literature, node by node

Parkinson's spine

The mitochondrial-dysfunction → dopaminergic-degeneration edge we reconstruct is the

Nakamura

lab's own work (Berthet 2014; CHCHD2, Sci Adv 2025); the parkinsonian-deficit outcome is the

Kreitzer

lab's basal-ganglia circuit (Kravitz 2010, Nature).

Alzheimer's arm

Our neuroinflammation + BBB node is

Akassoglou's

fibrin→microglia mechanism (Merlini 2019; Mendiola 2023); the APOE→microglia node is

Huang's

APOE4 work (2024-25); TREM2→network is

Mucke

(Das 2021).

The field owns the endogenous mechanism; Aitiome adds the exposome layer.

What we deliberately do not use

- **Bioactivity as a discriminator** (ToxCast/Tox21 hitcalls, GenRA, DeepTox) - anti-diagnostic here; ToxCast covers only ~40% neural-relevant targets (Mack 2024).
- **Circular knowledge graphs** (ComptoxAI, AlzKB, PrimeKG, Hetionet) - edges are CTD/AOP-derived or drug-keyed.
- **Structure / QSAR & morphology** - general similarity is not neurotoxicity.
- **CTD inferred associations** - inference by study volume (acetaminophen: 80 inferred PD links). Only curated DirectEvidence counts.

Anything that smuggles general bioactivity back in, or is circular with our own curation, is disqualified.

Every claim links to its primary source

Aitiome

exposome / neurodegeneration

engine live

Light

disease axis

Parkinson's

Alzheimer's

AOP-3 endorsed anchor

OECD-endorsed AOP-3 spine (complex-I -> mitochondrial dysfunction -> nigrostriatal degeneration -> parkinsonian deficits). The validated anchor.

The engine reasons using over 100,000,000 sources, each evidence marker in a synthetic

links to the matching source below, and every entry links to the original material.

1

Comparative Toxicogenomics Database (CTD)

DIAGNOSTIC

Davis AP, Wiegiers TC, Johnson RJ, et al. Comparative Toxicogenomics Database (CTD): update 2023. Nucleic Acids Research. 2023;51(D1):D1257-D1262.

https://ctdbase.org/

2

OECD Outcome Pathway Wiki (AOP-3)

DIAGNOSTIC

OECD Outcome Pathway Wiki. AOP 3: Inhibition of the mitochondrial complex I of nigro-striatal neurons leads to parkinsonian motor deficits (WPHA/WNT Endorsed).

https://aopwiki.org/aops/3

EPA ToxCast / invitroDB via NICEATM ICE

CORROBORATION

U.S. EPA ToxCast/Tox21 invitroDB (v4.2), accessed through the NICEATM Integrated Chemical Environment (ICE). Active hitcalls only; queried on the salt-form-correct record.

https://ice.ntp.niehs.nih.gov/

MitoCarta3.0

GROUNDING

Rath S, Sharma R, Gupta R, et al. MitoCarta3.0: an updated mitochondrial proteome now with sub-organelle localization and pathway annotations. Nucleic Acids Research. 2021;49(D1):D1541-D1547.

https://www.broadinstitute.org/mitocarta

Kamath 2022 (SN dopaminergic atlas)

GROUNDING

Kamath T, Abdullaouf A, Burris SJ, et al. Single-cell genomic profiling of human dopamine neurons identifies a population that selectively degenerates in Parkinson's disease. Nature Neuroscience. 2022;25(5):588-595.

https://doi.org/10.1038/s41593-022-01061-1

FDA FAERS (openFDA)

CORROBORATION

U.S. FDA Adverse Event Reporting System (FAERS), via the openFDA drug/event API. Disproportionality (ROR) for the parkinsonism MedDRA cluster.

https://open.fda.gov/apis/drug/event/

Human epidemiology (published cohorts / meta-analyses)

CORROBORATION

Quantified PD/AD exposure risk curated from published cohorts and meta-analyses (e.g., Tanner et al. 2011, Environmental Health Perspectives; Pezzoli & Cereda 2013, Neurology).

https://pubmed.ncbi.nlm.nih.gov/

B3DB + BOILED-Egg (brain exposure)

CORROBORATION

Meng F, Xi Y, Huang J, Ayers JW. A curated diverse molecular database of blood-brain barrier permeability with chemical descriptors (B3DB). Scientific Data. 2021;8:289. Daina A, Zoete V. A BOILED-Egg to predict gastrointestinal absorption and brain access of small molecules. ChemMedChem. 2016;11(11):1117-1121.

https://github.com/theochem/B3DB

EPA CompTox Chemicals Dashboard / DSSTox

IDENTITY

U.S. EPA CompTox Chemicals Dashboard (DSSTox). DTXSID-first identity resolution with the salt-form-correct registered substance.

https://comptox.epa.gov/dashboard

ONE ENGINE, TWO INTERFACES

An external agent queries the same engine

1

Nine primary sources, each a resolvable citation

2

Tagged by role: diagnostic, corroboration, grounding

A scientist and an agent query the same tools

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exposome / neurodegeneration

engine live

Light

disease axis

Parkinson's

Alzheimer's

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The validated anchor.

ONE ENGINE, TWO INTERFACES

An external agent queries the same engine

MCP tools

assess_compound

resolve, grade, reconstruct, trace

synthesis, assessment

calibrated, cited

run_validation

recover 12 / reject 15, fp=fn=0

get_pathway

endorsed AOP, grounded

discovery_map

the negative-results map

benchmark

the falsification (bioactivity AUROC vs decoys)

sources

research-style citations with links

resolve_compound

DTXSID-first, salt-form correct

example call

{

"tool": "assess_compound",

"input": {

"id": "rotenone"

},

"output": {

"call": "positive",

"tier": "assay_mechanism_recovered",

"rationale": "Curated convergence: CTD Parkinson's DirectEvidence AND registered neuro-AOP stressor."

}

}

1

Eight MCP tools - the same engine, exposed to agents

2

An agent gets the same graded, cited call

BUILT WITH CLAUDE, END TO END

Claude Code builds it. The API reasons. Claude Science verifies.

Claude Code

Built the whole system - two parallel streams (engine and visualization) over a versioned contract, the deterministic core, the falsification harness, the hero, the deploy.

Claude API (Opus 4.8)

The in-app evidence-reasoner (explains, never decides) and a command-line curation agent that assembles curated evidence for a new chemical with web tools.

Claude Science

The evidence-assembly and **verification** layer: it checks a draft against the primary sources and produces a reproducible artifact, turning a hypothesis into a curated call.

A full-stack Claude pipeline - and Claude still never enters the decision path.

Deterministic core, auditable end to end

- One Go binary embeds the curated data; the recovery decision is **deterministic** - no LLM in the decision path.
- Transport-agnostic service, thin HTTP + MCP adapters - a scientist and an external agent query the same tools.
- DTXSID-first, salt-form-correct identity (the paraquat-dichloride trap, tested).
- Every claim links to the original source (CTD, AOP-Wiki, MitoCarta, Kamath, ToxCast/ICE, FAERS, B3DB).
- **Beyond the 27:** Claude Science assembles curated evidence, POST /assess-curated grades it - an unverified draft is a hypothesis, never a recovery.
- ~13 MB container, live on fly.io. Reproducible: same input, same output.

Challenges, and what they taught us

- **Novelty was the wrong optimization.** Five prior-art scans found the whole design space already published - so we built the validated prototype the field called for, and positioned it accurately.
- **Bioactivity is anti-diagnostic.** The deepest surprise: on our own data it scores worse than chance against the decoys. Activity cannot be the discriminator.
- **Identity correctness is day one.** The paraquat salt-form trap silently returns "no data" and corrupts everything downstream.
- **Our own test harness caught our own overclaim.** A red-team test we wrote failed, and forced the stronger, evidence-based statement. Ground claims in executable evidence.

What this hackathon produced

A result that holds

Recovery works (13/13 PD; 12/12 AD), specificity is proven by falsification (bioactivity at or below chance), and the discovery limits are mapped, not hidden.

Calibration as the edge

For a mechanism-first audience, a calibrated, auditable engine beats an overclaimed watchlist. Every claim is cited to its source.

Shipped, live, reproducible

Built end to end on Claude, deterministic where it must be, running at aitiome.fly.dev with an MCP interface for agents.

Limitations and next steps

- A curated benchmark with an adversarial control (27), not a population generalization study.
- Recovery shares curated provenance with the labels - defended by the independent-source ablation and the orthogonal negatives; it is a sanity check, not the headline.
- Anchored on the Parkinson's / mitochondrial route; Alzheimer's is a scaffold extension.
- **Next:** a larger, blinded neural-specific reference set to power the one qualified lead (AUROC 0.72, currently perm-p 0.155).
- **Next:** run the bounded Boltz-2 Q-site benchmark - physics, not annotation or activity.
- **Next:** extend the endorsed scaffold to an Alzheimer's hallmark.



Less flashy than the original pitch. Far more trustworthy.

Validated recovery, adversarial specificity proven by falsification, and a mapped account of the limits - for a mechanism-first audience, that is the point.

`aitiome.fly.dev` / `github.com/jonradoff/aitiome`